```sql
/* ============================================================
   LAB: AI-Based Telecom Customer Churn Analytics
   Database: MSSQL Server
   Use Case: Telecom Customer Churn Prediction
   ============================================================ */

-- 1. Create Database
CREATE DATABASE TelecomAI_DB;

USE TelecomAI_DB;

/* ============================================================
   2. Create Customer Table
   ============================================================ */

CREATE TABLE telecom_customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    gender VARCHAR(10),
    age INT,
    contract_type VARCHAR(20),
    monthly_bill DECIMAL(10,2),
    internet_usage_gb DECIMAL(10,2),
    call_minutes INT,
    complaint_count INT,
    payment_delay_days INT,
    churn_status VARCHAR(10)
);

/* ============================================================
   3. Insert Sample Telecom Data
   ============================================================ */

INSERT INTO telecom_customers
(customer_id, customer_name, gender, age, contract_type, monthly_bill,
 internet_usage_gb, call_minutes, complaint_count, payment_delay_days, churn_status)
VALUES
(1, 'Aung Aung', 'Male', 28, 'Prepaid', 25000, 35.5, 400, 3, 10, 'Yes'),
(2, 'Su Su', 'Female', 32, 'Postpaid', 45000, 80.0, 900, 0, 0, 'No'),
(3, 'Kyaw Kyaw', 'Male', 45, 'Prepaid', 18000, 20.0, 250, 4, 15, 'Yes'),
(4, 'Mya Mya', 'Female', 25, 'Postpaid', 60000, 120.5, 1200, 1, 2, 'No'),
(5, 'Hla Hla', 'Female', 39, 'Prepaid', 22000, 25.5, 300, 5, 20, 'Yes'),
(6, 'Tun Tun', 'Male', 30, 'Postpaid', 52000, 100.0, 1000, 0, 0, 'No'),
(7, 'Nandar', 'Female', 27, 'Prepaid', 20000, 18.5, 200, 2, 8, 'Yes'),
(8, 'Moe Moe', 'Female', 35, 'Postpaid', 48000, 75.0, 850, 1, 1, 'No'),
(9, 'Zaw Zaw', 'Male', 42, 'Prepaid', 26000, 30.0, 350, 4, 12, 'Yes'),
(10, 'Ei Ei', 'Female', 29, 'Postpaid', 55000, 110.0, 1100, 0, 0, 'No');
```

Day-8 Assignment_churnprediction

DE_2026_0094

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```sql
/* =========================================================
   4. Data Exploration
   --======================================================= */

-- View all data
SELECT * FROM telecom_customers;
```

121 % ⊗ 2 ⚠ 0 Ln: 13, Ch: 1 MIXED CRLF UTF-8

Results | Messages

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes
2	2	Su Su	Female	32	Postpaid	45000.00	80.00	900	0	0	No
3	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes
4	4	Mya Mya	Female	25	Postpaid	60000.00	120.50	1200	1	2	No
5	5	Hla Hla	Female	39	Prepaid	22000.00	25.50	300	5	20	Yes
6	6	Tun Tun	Male	30	Postpaid	52000.00	100.00	1000	0	0	No
7	7	Nandar	Female	27	Prepaid	20000.00	18.50	200	2	8	Yes
8	8	Moe Moe	Female	35	Postpaid	48000.00	75.00	850	1	1	No

```sql
-- Count churn and non-churn customers
--SELECT churn_status   - Retrieves churn category
--COUNT(*)  - Counts rows
--AS total_customers    - Gives alias name
--FROM telecom_customers     - Reads from telecom table
--GROUP BY churn_status - Groups rows by churn type


SELECT
    churn_status,
    COUNT(*) AS total_customers
FROM telecom_customers
GROUP BY churn_status;
```

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Results | Messages

	churn_status	total_customers
1	No	5
2	Yes	5

```sql
-- Average monthly bill by churn status
--AVG(monthly_bill) -Calculates average bill
--AS avg_monthly_bill    -Renames output
--GROUP BY churn_status -Separates churn vs non-churn

SELECT
    churn_status,
    AVG(monthly_bill) AS avg_monthly_bill
FROM telecom_customers
GROUP BY churn_status;
```

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Results | Messages

	churn_status	avg_monthly_bill
1	No	52000.000000
2	Yes	22200.000000

```sql
-- Average complaints by churn status
SELECT
    churn_status,
    AVG(complaint_count) AS avg_complaints
FROM telecom_customers
GROUP BY churn_status;
```

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Results | Messages

	churn_status	avg_complaints
1	No	0
2	Yes	3

Day-8 Assignment_churnprediction

DE_2026_0094

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```sql
   5. Data Preparation
   Convert Churn Status into Numeric Label
   Yes = 1, No = 0
   ======================================================= */
    --CASE  - Conditional logic
    --WHEN churn_status = 'Yes' - Checks churn condition
    --THEN 1    - Assigns numeric value
    --ELSE 0    - Assigns non-churn value
    --END AS churn_label    - Creates new column

SELECT
    customer_id,
    customer_name,
    age,
    contract_type,
    monthly_bill,
    internet_usage_gb,
    call_minutes,
    complaint_count,
    payment_delay_days,
    CASE
        WHEN churn_status = 'Yes' THEN 1
        ELSE 0
    END AS churn_label
FROM telecom_customers;
```

91 % | 2 | 0 | Ln: 117, Ch: 24 | MIXED | CRLF | UTF-8

Results | Messages

	customer_id	customer_name	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_label
1	1	Aung Aung	28	Prepaid	25000.00	35.50	400	3	10	1
2	2	Su Su	32	Postpaid	45000.00	80.00	900	0	0	0
3	3	Kyaw Kyaw	45	Prepaid	18000.00	20.00	250	4	15	1
4	4	Mya Mya	25	Postpaid	60000.00	120.50	1200	1	2	0
5	5	Hla Hla	39	Prepaid	22000.00	25.50	300	5	20	1
6	6	Tun Tun	30	Postpaid	52000.00	100.00	1000	0	0	0
7	7	Nandar	27	Prepaid	20000.00	18.50	200	2	8	1
8	8	Moe Moe	35	Postpaid	48000.00	75.00	850	1	1	0

```sql
/* =======================================================
   6. Simple AI Rule-Based Churn Risk Scoring
   This simulates AI scoring logic inside SQL.
   ======================================================= */
 --  Condition 1
   --WHEN complaint_count >= 4

   --Customers with many complaints may leave.
   --Condition 2
   --OR payment_delay_days >= 15

   --Late payments indicate dissatisfaction or financial issues.
   --Condition 3
   --OR contract_type = 'Prepaid'
   --Prepaid users are easier to switch to competitors.

SELECT
    customer_id,
    customer_name,
    contract_type,
    monthly_bill,
    complaint_count,
    payment_delay_days,
    churn_status,

    CASE
        WHEN complaint_count >= 4
             OR payment_delay_days >= 15
             OR contract_type = 'Prepaid'
        THEN 'High Risk'

        WHEN complaint_count BETWEEN 2 AND 3
             OR payment_delay_days BETWEEN 7 AND 14
        THEN 'Medium Risk'

        ELSE 'Low Risk'
    END AS predicted_churn_risk

FROM telecom_customers;
GO
```

56 % | 2 | 0 | Ln: 137, Ch: 19 | MIXED | CRLF | UTF-8

Results | Messages

	customer_id	customer_name	contract_type	monthly_bill	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
3	3	Kyaw Kyaw	Prepaid	18000.00	4	15	Yes	High Risk
4	4	Mya Mya	Postpaid	60000.00	1	2	No	Low Risk
5	5	Hla Hla	Prepaid	22000.00	5	20	Yes	High Risk
6	6	Tun Tun	Postpaid	52000.00	0	0	No	Low Risk
7	7	Nandar	Prepaid	20000.00	2	8	Yes	High Risk
8	8	Moe Moe	Postpaid	48000.00	1	1	No	Low Risk
9	9	Zaw Zaw	Prepaid	26000.00	4	12	Yes	High Risk
10	10	Ei Ei	Postpaid	55000.00	0	0	No	Low Risk

```sql
/* =======================================================
   7. Create View for Churn Risk Dashboard
   ======================================================= */

CREATE VIEW vw_customer_churn_risk AS
SELECT
    customer_id,
    customer_name,
    gender,
    age,
    contract_type,
    monthly_bill,
    internet_usage_gb,
    call_minutes,
    complaint_count,
    payment_delay_days,
    churn_status,

    CASE
        WHEN complaint_count >= 4
             OR payment_delay_days >= 15
             OR contract_type = 'Prepaid'
        THEN 'High Risk'

        WHEN complaint_count BETWEEN 2 AND 3
             OR payment_delay_days BETWEEN 7 AND 14
        THEN 'Medium Risk'

        ELSE 'Low Risk'
    END AS predicted_churn_risk
FROM telecom_customers;
GO
```

62 % | 2 | 0 | Ln: 153, Ch: 9 | MIXED | CRLF | UTF-8

Results | Messages

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes	High Risk
2	2	Su Su	Female	32	Postpaid	45000.00	80.00	900	0	0	No	Low Risk
3	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes	High Risk
4	4	Mya Mya	Female	25	Postpaid	60000.00	120.50	1200	1	2	No	Low Risk
5	5	Hla Hla	Female	39	Prepaid	22000.00	25.50	300	5	20	Yes	High Risk
6	6	Tun Tun	Male	30	Postpaid	52000.00	100.00	1000	0	0	No	Low Risk
7	7	Nandar	Female	27	Prepaid	20000.00	18.50	200	2	8	Yes	High Risk
8	8	Moe Moe	Female	35	Postpaid	48000.00	75.00	850	1	1	No	Low Risk

Day-8 Assignment_churnprediction

DE_2026_0094

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```sql
/* ========================================================
   8. Query the Churn Risk Dashboard View
   ======================================================== */

SELECT * FROM vw_customer_churn_risk;

-- Show only high-risk customers
SELECT *
FROM vw_customer_churn_risk
WHERE predicted_churn_risk = 'High Risk';

/* ========================================================
```

100 % | 2 | 0 | Ln: 153, Ch: 9 | MIXED | CRLF | UTF-8

Results | Messages

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes	High Risk
2	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes	High Risk
3	5	Hla Hla	Fem...	39	Prepaid	22000.00	25.50	300	5	20	Yes	High Risk
4	7	Nandar	Fem...	27	Prepaid	20000.00	18.50	200	2	8	Yes	High Risk
5	9	Zaw Zaw	Male	42	Prepaid	26000.00	30.00	350	4	12	Yes	High Risk

```sql
/* ========================================================
   9. Retention Campaign Recommendation
   ======================================================== */

SELECT
    customer_id,
    customer_name,
    predicted_churn_risk,
    CASE
        WHEN predicted_churn_risk = 'High Risk'
        THEN 'Offer discount package and priority customer support'

        WHEN predicted_churn_risk = 'Medium Risk'
        THEN 'Send loyalty offer and service improvement message'

        ELSE 'Maintain normal service communication'
    END AS recommended_action
FROM vw_customer_churn_risk;
GO
```

100 % | 2 | 0 | Ln: 153, Ch: 9 | MIXED | CRLF | UTF-8

Results | Messages

	customer_id	customer_name	predicted_churn_risk	recommended_action
3	3	Kyaw Kyaw	High Risk	Offer discount package and priority customer sup...
4	4	Mya Mya	Low Risk	Maintain normal service communication
5	5	Hla Hla	High Risk	Offer discount package and priority customer sup...
6	6	Tun Tun	Low Risk	Maintain normal service communication
7	7	Nandar	High Risk	Offer discount package and priority customer sup...
8	8	Moe Moe	Low Risk	Maintain normal service communication
9	9	Zaw Zaw	High Risk	Offer discount package and priority customer sup...
10	10	Ei Ei	Low Risk	Maintain normal service communication

```sql
/* ========================================================
   10. Model Evaluation Simulation
   Compare Actual Churn Status with Predicted Risk
   ======================================================== */

SELECT
    churn_status,
    predicted_churn_risk,
    COUNT(*) AS total_customers
FROM vw_customer_churn_risk
GROUP BY churn_status, predicted_churn_risk;
GO
```

100 % | 2 | 0 | Ln: 228, Ch: 4 | MIXED | CRLF | UTF-8

Results | Messages

	churn_status	predicted_churn_risk	total_customers
1	Yes	High Risk	5
2	No	Low Risk	5

Query executed successfully. | localhost\SQLEXPRESS (15.0 ... | LINLAEPHYU\Asus (53) | TelecomAI_DB | 00:00:01 | Row: 1, Col: 1 | 10 rows